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Experiment-24

Question

Write a C program that displays a menu and uses a **switch statement** to print the selected **Decreasing Pattern**. Use **nested loops without arrays**. Each pattern may be implemented in a separate **parameterless function with void return type**.

Menu

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Algorithm

1. Start.
2. Declare the required variables.
3. Declare four parameterless functions with `void` return type:
 - o `pattern1()`
 - o `pattern2()`
 - o `pattern3()`
 - o `pattern4()`
4. Display the menu containing the four decreasing patterns and Exit option.
5. Input the user's choice.
6. Use a `switch` statement to perform the selected operation.
7. If choice is **1**, call `pattern1()`.
 - o Input number of rows `n`.
 - o Print `n` asterisks in the first row.
 - o Decrease the number of asterisks by one in every successive row.
8. If choice is **2**, call `pattern2()`.
 - o Input `n`.
 - o Print natural numbers from `1` to `n` in the first row.
 - o Reduce the number of values by one in every successive row.
9. If choice is **3**, call `pattern3()`.
 - o Input `n`.
 - o Start with the value `n`.
 - o Print the value as many times as its value.

- Decrease the value by one in every successive row.
- 10. If choice is **4**, call `pattern4()`.
 - Input `n`.
 - Print numbers from `n` down to `1` in the first row.
 - Start each successive row with one smaller value.
- 11. If choice is **5**, exit the program.
- 12. For any other choice, display **Invalid choice**.
- 13. Repeat the menu until the user selects Exit.
- 14. Stop.

Test-Case Table

Test Case	Choice	Input	Expected Output	Results
1	1	4	Decreasing asterisk triangle with 4 rows	Pass
2	2	4	1 2 3 4, 1 2 3, 1 2, 1	Pass
3	3	4	4 4 4 4, 3 3 3, 2 2, 1	Pass
4	4	4	4 3 2 1, 3 2 1, 2 1, 1	Pass
5	1	1	Single *	Pass
6	2	0	Invalid number of rows	Pass
7	6	-	Invalid choice	Pass
8	5	-	Program exits	Pass

C Program

```
#include <stdio.h>
```

```
void pattern1(void);
void pattern2(void);
void pattern3(void);
void pattern4(void);
```

```
int main()
{
    int choice;

    do
    {
        printf("\n--- Decreasing Patterns Menu ---\n");
        printf("1. Decreasing right-angled triangle\n");
        printf("2. Decreasing natural-number pattern\n");
        printf("3. Decreasing repeated-number pattern\n");
```

```

printf("4. Reverse-number triangle\n");
printf("5. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        pattern1();
        break;

    case 2:
        pattern2();
        break;

    case 3:
        pattern3();
        break;

    case 4:
        pattern4();
        break;

    case 5:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice.\n");
}

} while(choice != 5);

return 0;
}

/* Decreasing right-angled triangle */
void pattern1(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

```

```

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= i; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

/* Decreasing natural-number pattern */
void pattern2(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

/* Decreasing repeated-number pattern */
void pattern3(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = n; i >= 1; i--)

```

```

    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

/* Reverse-number triangle */
void pattern4(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = n; i >= 1; i--)
    {
        for(j = i; j >= 1; j--)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

```

Compilation and Execution

```
(kali㉿kali)-[~]  
$ vi hdoc25.c  
  
(kali㉿kali)-[~]  
$ gcc hdoc25.c -o hdoc25  
  
(kali㉿kali)-[~]  
$ ./hdoc25
```

```
— Decreasing Patterns Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 1  
Enter number of rows: 4  
* * * *  
* * *  
* *  
*  
  
— Decreasing Patterns Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 2
```

```
Enter number of rows: 4  
1 2 3 4  
1 2 3  
1 2  
1  
  
— Decreasing Patterns Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 3  
Enter number of rows: 4  
4 4 4 4  
3 3 3  
2 2  
1  
  
— Decreasing Patterns Menu —  
1. Decreasing right-angled triangle  
2. Decreasing natural-number pattern  
3. Decreasing repeated-number pattern  
4. Reverse-number triangle  
5. Exit  
Enter your choice: 4  
Enter number of rows: 4  
4 3 2 1  
3 2 1
```

```
3 2 1
2 1
1
```

— Decreasing Patterns Menu —

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 1

Enter number of rows: 1

*

— Decreasing Patterns Menu —

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 2

Enter number of rows: 0

Invalid number of rows.

— Decreasing Patterns Menu —

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

4. Reverse-number triangle

5. Exit

Enter your choice: 6

Invalid choice.

— Decreasing Patterns Menu —

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 5

Exiting program...